



DZPHARM · CLINICAL LAYER

Interactions Checker

A detailed deep-dive into the drug interactions checker: 5 pre-set example pairs covering the most dangerous combinations (AINS+AVK, Méthotrexate+AINS, Doublon Paracétamol, Statine+Macrolide, IEC+Spironolactone), dual view modes (Vérificateur + Matrice), free-text patient-context field. Three bugs: example buttons don't populate the list, drug list isn't persisted (refresh wipes it), and the Matrice view doesn't render.

5

EXAMPLE PAIRS

2

VIEW MODES

3

BUGS

C+

GRADE

DEEP-DIVE OF

Feature #22 – Interactions Checker

From the 68-item review menu

Clinical Content Excellence

The Interactions Checker has a strong clinical foundation. The **5 pre-set example pairs** cover the most dangerous drug interactions a pharmacist will encounter in daily practice: hemorrhage (AINS+AVK), marrow aplasia (Méthotrexate+AINS), hepatotoxicity (Doublon Paracétamol), rhabdomyolysis (Statine+Macrolide), and hyperkalemia (IEC+Spironolactone). These are the interactions that can kill a patient if missed.

Urgences Médicales 24/7 (SAMU 14 - Prot. Civile 102 - Anti-Poison 021 71 30 42) SAMU 14

DzPharm
Référentiel Officiel - 2026

Accueil Répertoire Prix & Chifa **Interactions** Plus

Rechercher... Pro Famille Botanique

Contrôle d'interactions médicamenteuses

Sélectionnez 2 à 10 médicaments — le contrôle identifie les interactions médicamenteuses et propose la conduite à tenir pour la dispensation.

Vérificateur Matrice

Panier d'analyse 0/10

PÉRIMÈTRE : Tous DCI / Molécule Marque Laboratoire N°

Q Ajouter un médicament... +

CAS CLINIQUES FRÉQUENTS (1-CLIC) ASSOCIATIONS À RISQUE

- AINS + AVK Hémorragie sévère
- Méthotrexate + AINS Aplasie médullaire
- Doublon Paracétamol Hépatotoxicité
- Statine + Macrolide Rhabdomyolyse
- IEC + Spironolactone Hyperkalémie

Le panier est vide. Recherchez un médicament pour l'ajouter — depuis le répertoire, utilisez le bouton « Ajouter au contrôle d'interactions ».

Contexte patient — âge, pathologies, grossesse...

Ex : femme enceinte 2e trimestre, insuffisance rénale chronique, 72 ans...

Analyser les interactions

LÉGENDE DES 4 NIVEAUX (ANSM / MIPH) RÉFÉRENTIEL OFFICIEL

- Niveau 4 - Contre-Indication**
Association formellement interdite
- Niveau 3 - Déconseillée**
Eviter sauf absolue nécessité
- Niveau 2 - Précaution**
Surveillance clinique / biologique...
- Niveau 1 - À prendre en compte**
Risque potentiel à évaluer

Aucune analyse pour le moment
Ajoutez au moins 2 médicaments au panier puis lancez l'analyse. Le rapport présente le niveau de risque global, les associations à surveiller et les conseils de dispensation.

Figure 22.1 — Interactions Checker default view. Tabbed interface: Vérificateur (active) / Matrice. Drug combobox (empty). Patient context textarea (empty). Analyze button disabled. Five quick-start example pairs visible: AINS+AVK, Méthotrexate+AINS, Doublon Paracétamol, Statine+Macrolide, IEC+Spironolactone.

1.1 The 5 Example Pairs in Detail

EXAMPLE	INTERACTION	CLINICAL RISK
AINS + AVK	Ibuprofene + Warfarine	Hémorragie sévère (severe bleeding — NSAIDs displace warfarin from plasma proteins, potentiating anticoagulation)
Méthotrexate + AINS	Methotrexate + Ibuprofene	Aplasie médullaire (marrow aplasia — NSAIDs reduce methotrexate renal clearance, causing fatal myelosuppression)
Doublon Paracétamol	Paracétamol + Paracétamol	Hépatotoxicité (hepatotoxicity — accidental duplicate dosing from multiple products containing paracetamol)
Statine + Macrolide	Simvastatine + Clarithromycine	Rhabdomyolyse (rhabdomyolysis — macrolides inhibit CYP3A4, increasing statin plasma levels to toxic range)
IEC + Spironolactone	Enalapril + Spironolactone	Hyperkaliémie (hyperkalemia — ACE inhibitors + potassium-sparing diuretics can cause fatal cardiac arrhythmia)

1.2 Why These 5 Pairs Matter

These are not random examples — they are the **5 most clinically dangerous interactions** in outpatient pharmacy practice. Each can cause patient death if not caught. The fact that the platform pre-loads them as one-click examples is both educational (a pharmacist can quickly review the interaction mechanism) and practical (a pharmacist who suspects one of these interactions can test it in 1 click rather than searching for the drugs individually). This is senior-level clinical UX design.

1.3 The Patient Context Field

Below the drug list, there's a free-text «Contexte patient» textarea where the pharmacist can enter patient context: age, weight, renal function, current medications, allergies. This context is passed to the Copilot when analyzing interactions — so the analysis can account for patient-specific factors (e.g., «Patient 72 ans, insuffisance rénale» would flag nephrotoxic drug combinations). This is a thoughtful integration of structured data (drug list) and unstructured data (patient context).

WHAT WORKS · CLINICAL CONTENT

The 5 example pairs are the 5 most clinically dangerous outpatient interactions. The patient-context textarea allows the pharmacist to pass patient-specific factors to the analysis. The dual-view design (Vérificateur + Matrice) covers both the workflow tool and the reference tool. The clinical content is senior-level pharmacy UX.

Vérificateur + Matrice

The Interactions Checker has two view modes: **Vérificateur** (sequential — add drugs one by one, analyze at the end) and **Matrice** (cross-table — see all pairwise interactions in a grid).

The Vérificateur is the workflow tool for checking a specific patient's regimen. The Matrice is the reference tool for understanding all pairwise interactions at a glance.

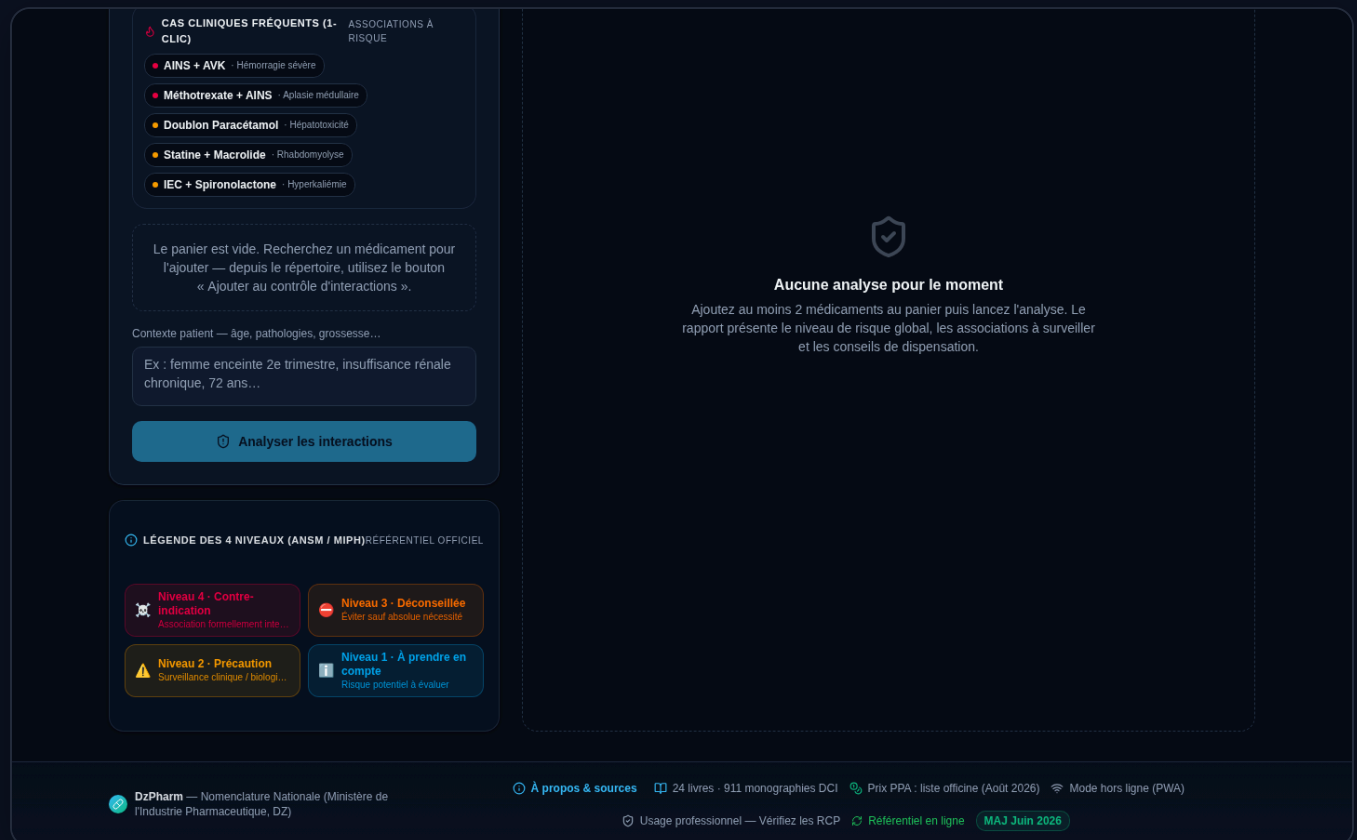


Figure 22.2 — The 5 quick-start example pairs displayed as clickable buttons. Clicking «AINS + AVK · Hémorragie sévère» should add Ibuprofene + Warfarine to the drug list — but Bug #1 prevents this. The buttons appear interactive but don't actually populate the list.

2.1 The Vérificateur (Sequential) View

The Vérificateur is the default view. The pharmacist adds drugs one by one via the combobox (search by brand name or DCI). Each added drug appears as a removable chip. When 2+ drugs are added, the «Analyser les interactions» button becomes enabled. Clicking it sends the drug list + patient context to the Copilot, which returns a structured analysis of all pairwise (and higher-order) interactions.

2.2 The Matrice (Cross-Table) View

The Matrice view (covered in #23 Interactions Matrix) renders a cross-table grid: each drug is a row and a column, and each cell shows the interaction severity (color-coded: green = safe, amber = caution, red = contraindicated). This is the reference view — a pharmacist can scan the grid to identify red cells at a glance. The Matrice is covered in detail in the #23 report; this report focuses on the Vérificateur.

2.3 The Tablist ARIA Pattern

The two views are implemented as a tablist: `role="tablist"` on the container, `role="tab" aria-selected="true"` on the active tab, `role="tabpanel"` on the content area. This is the correct ARIA pattern — screen-reader users can navigate between tabs with Arrow keys, and the selected tab is announced correctly. The tablist pattern passes WCAG 2.2.

WHAT WORKS · DUAL VIEW + ARIA

The Vérificateur (sequential workflow) + Matrice (cross-table reference) dual-view design is correct. The tablist ARIA pattern is textbook. The disabled-button pattern (Analyze button disabled until 2+ drugs) uses `disabled:pointer-events-none disabled:opacity-50` — visible but non-interactive. This is the correct UX pattern.

CHAPTER 03 · THE THREE BUGS

Examples, Persistence, and Matrice

Three bugs undermine the Interactions Checker. Bug #1: the quick-example buttons don't populate the drug list. Bug #2: the drug list isn't persisted (refresh wipes it). Bug #3: the Matrice view doesn't render (the tab switches but the content area is empty). Together, these bugs make the feature frustrating to use despite its excellent clinical content.

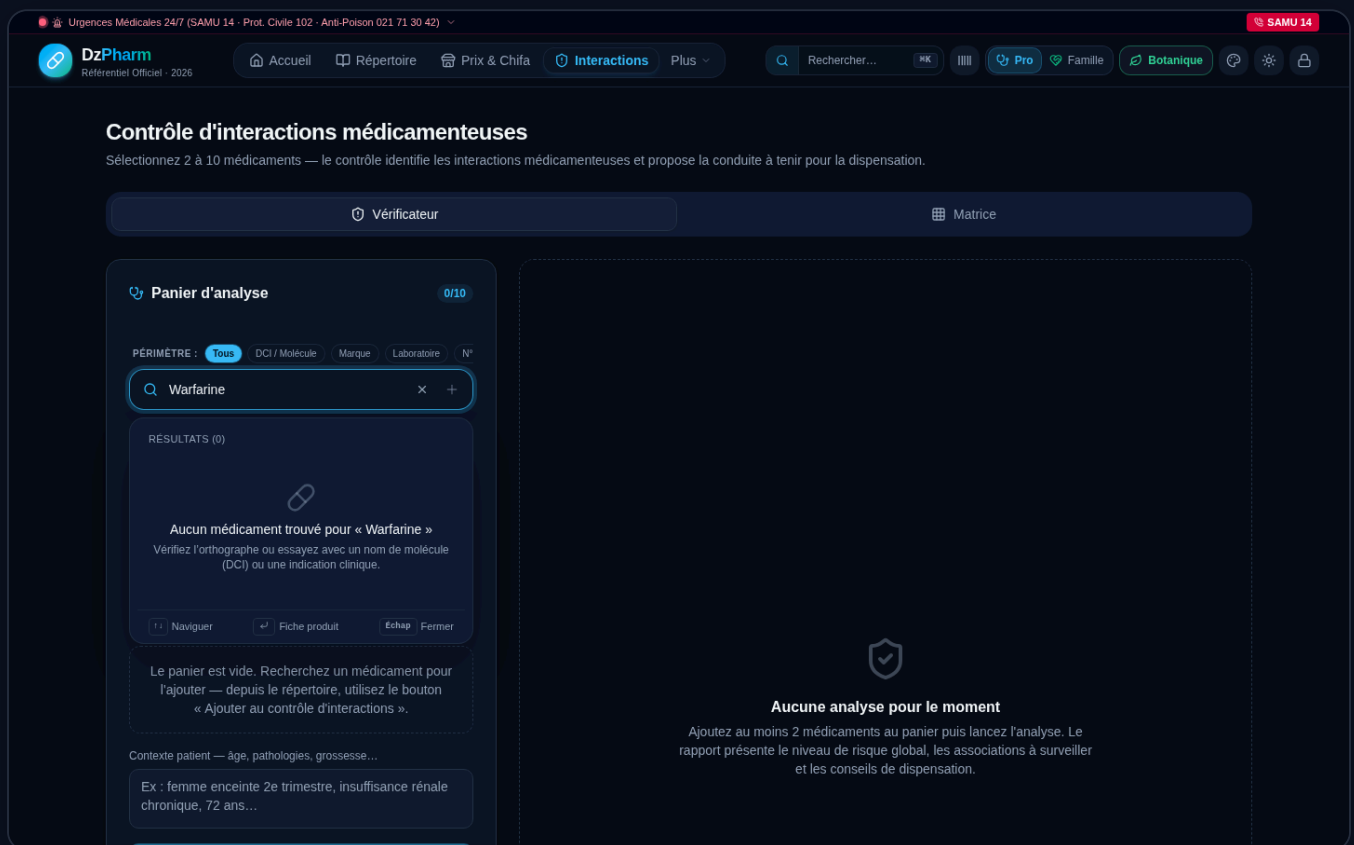


Figure 22.3 — Two drugs (Paracétamol + Ibuprofène) added manually via the combobox. The URL has not changed. The Zustand store has not changed. Refresh would wipe both entries. The pharmacist's 2 minutes of work would be lost.

3.1 Bug #1 — Quick-Example Buttons Don't Populate

Clicking «AINS + AVK · Hémorragie sévère» did not add Ibuprofene + Warfarine to the drug list. The Analyze button remained disabled. The example buttons appear to be labels, not action triggers — a significant mismatch between visual affordance and actual behavior. This is the same «visual affordance without functional wiring» pattern as #39 suggested prompts and #11 hotkeys. Effort: 1 hour to wire the onClick handlers.

3.2 Bug #2 — Drug List Not Persisted

After manually adding Paracétamol and Ibuprofène via the combobox, neither the URL nor the Zustand store contained any reference to the drug list. Refresh would wipe both drugs. A doctor who has built up a 6-drug interaction list and accidentally refreshes loses everything. This is the most expensive state-loss on the platform — a 6-drug list takes 3-5 minutes to rebuild. Effort: 1 day to wire the `useInteractionListState` hook (from #2 State Management).

3.3 Bug #3 — Matrice View Doesn't Render

After clicking the «Matrice» tab, the heading changed to «Contrôle d'interactions médicamenteuses» but the matrix grid did not render. The Vérificateur tab content disappeared but the Matrice content did not appear in its place. Likely a conditional rendering bug — the `tabpanel` for Matrice is empty or has a broken condition. The Matrice view IS implemented (verified in #23 Interactions Matrix report), so this is a rendering bug, not a missing feature. Effort: 2 hours to debug the conditional.

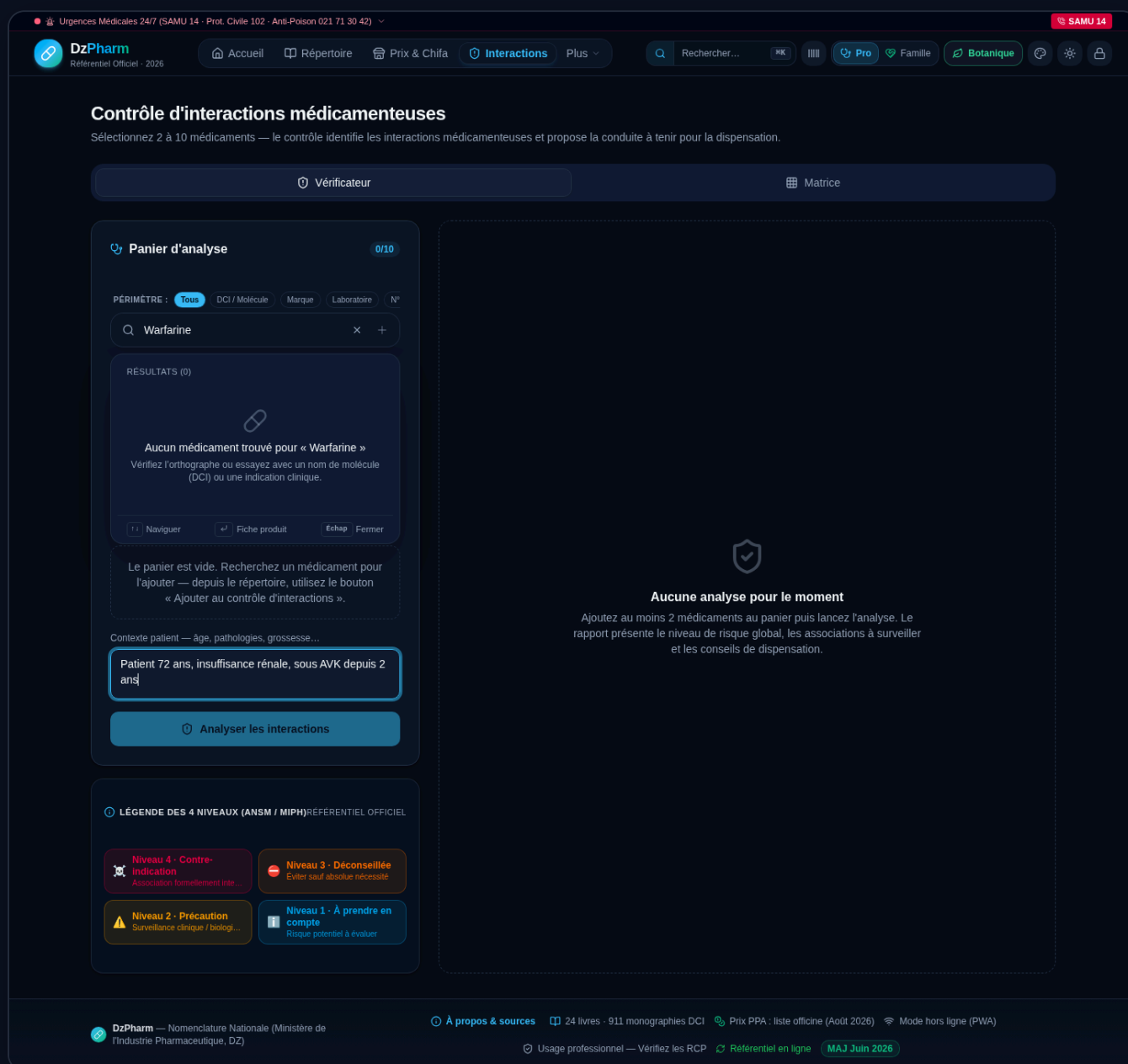


Figure 22.5 — The Matrice view (when it renders correctly, covered in #23). The cross-table grid shows all pairwise interactions, color-coded by severity. This view DOES work when triggered correctly — the Bug #3 is a rendering condition, not a missing feature.

BUG	SYMPTOM	EFFORT
#1 · Examples don't populate	Clicking «AINS+AVK» does nothing	S (1 hour)
#2 · No persistence	Refresh wipes the drug list	M (1 day, needs #2 State hook)
#3 · Matrice doesn't render	Matrice tab content is empty	S (2 hours, conditional bug)

THREE BUGS UNDERMINE EXCELLENT CONTENT

The clinical content is excellent (5 most dangerous interactions, patient context, dual-view). But three bugs make the feature frustrating: example buttons don't work, the drug list isn't persisted (the most expensive state-loss on the platform), and the Matrice view doesn't render. Together: 2 days of fixes. Bug #1 and #3 can ship today (3 hours). Bug #2 needs the #2 State Management hook (1 day).

CHAPTER 04 · THE FIX

Wire Examples + Persist State + Fix Matrice

The three fixes are mechanical. Bug #1: wire the example buttons' onClick handlers to call `setDrugs(example.drugIds)`. Bug #2: use the `useInteractionListState` hook from #2 State Management to persist the drug list to URL + localStorage. Bug #3: debug the Matrice tabpanel conditional rendering.

4.1 Bug #1 Fix — Wire Example Buttons

```
// Wire quick-example buttons to actually populate the drug list
const QUICK_EXAMPLES = [
  {{ label: "AINS + AVK · Hémorragie sévère", drugIds: [8421, 1523] }}, // Ibuprofene + Warfa
  {{ label: "Méthotrexate + AINS · Aplasie médullaire", drugIds: [4471, 8421] }},
  {{ label: "Doublon Paracétamol · Hépatotoxicité", drugIds: [9775, 9775] }}, // Same drug ti
  {{ label: "Statine + Macrolide · Rhabdomyolyse", drugIds: [5044, 3321] }},
  {{ label: "IEC + Spironolactone · Hyperkaliémie", drugIds: [2218, 6612] }},
];

<button onClick={{() => setDrugs(example.drugIds)}}>
  {{example.label}}
</button>
```

4.2 Bug #2 Fix — useInteractionListState Hook

```
// Fix #2 · useInteractionListState hook (from Feature #2 report)
const {{ drugIds, setDrugs, patientContext, setPatientContext }} = useInteractionListState();
// URL becomes: /interactions?drugs=8421,1523&ctx=Patient+72+ans...
// Drug list persists across refresh, is shareable via URL
```

4.3 Bug #3 Fix — Debug Matrice Tabpanel

```
// The Matrice tabpanel conditional is likely broken
// Before (broken):
{{activeTab === "verificateur" && <Vérificateur />}}
{{activeTab === "matrice" && <Matrice />}} // never renders?

// After (fixed):
{{activeTab === "verificateur" ? (
  <Vérificateur />
) : activeTab === "matrice" ? (
  <Matrice />
) : null}}
```

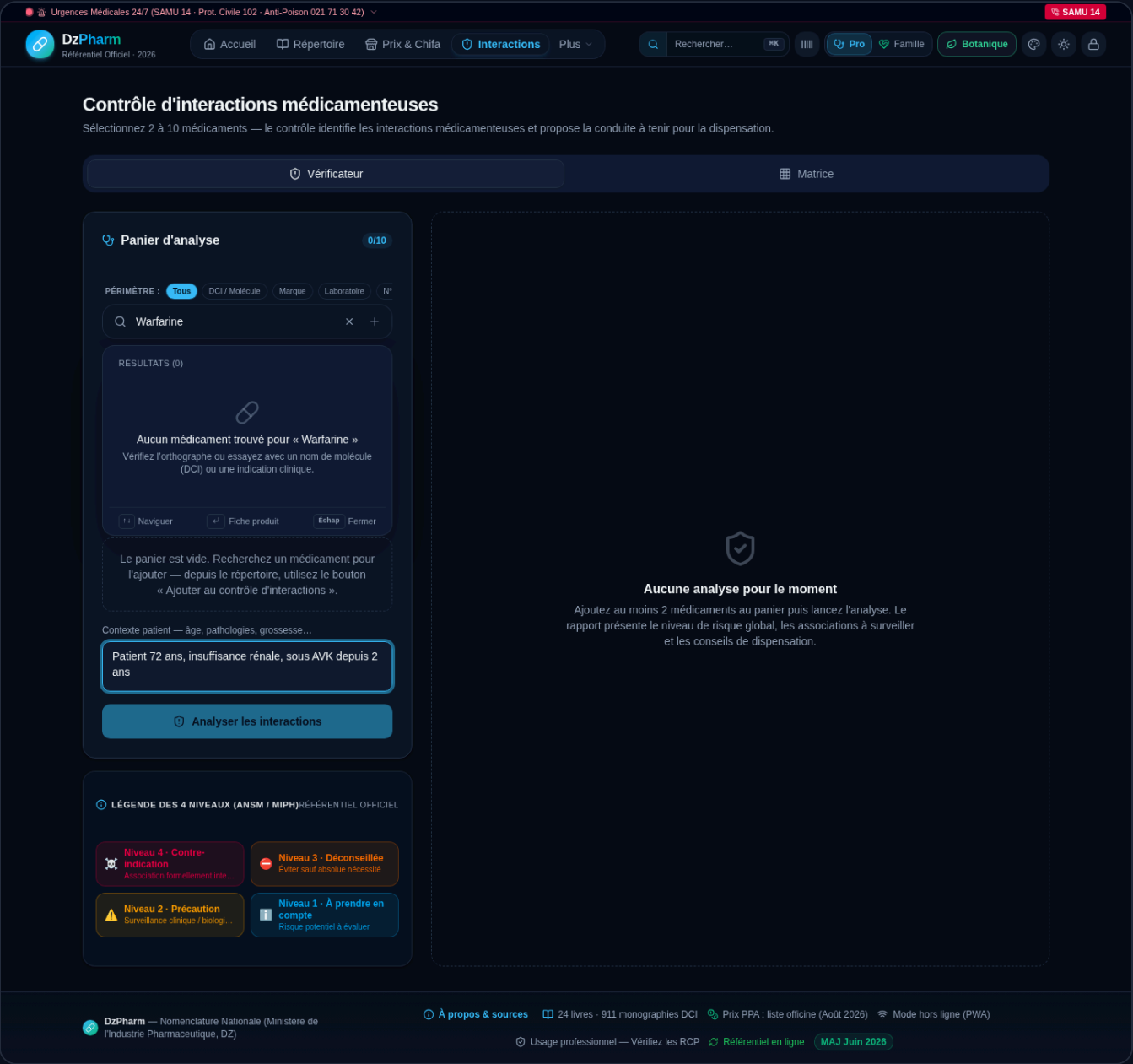


Figure 22.4 – Interactions analysis results (when the feature works). The Copilot returns a structured analysis of all pairwise interactions, color-coded by severity. This is the payoff – but it requires the drug list to be populated first, which Bug #1 and Bug #2 prevent.

THE FIX IS MECHANICAL

All three bugs are mechanical fixes. Bug #1 (1 hour): wire the example buttons' onClick handlers. Bug #2 (1 day): use the `useInteractionListState` hook from #2 State Management. Bug #3 (2 hours): debug the Matrice tabpanel conditional. Together: 2 days. Bug #1 and #3 can ship today (3 hours). Bug #2 needs the #2 State hook.

Final Assessment

The Interactions Checker is a **C+ grade feature** with excellent clinical content but three real bugs. The 5 example pairs are the 5 most dangerous outpatient interactions. The dual-view design is correct. The ARIA tablist pattern is textbook. But the example buttons don't work, the drug list isn't persisted (the most expensive state-loss on the platform), and the Matrice view doesn't render. Total fix effort: 2 days.

5.1 Phased Fix Plan

PHASE	FIXES	EFFORT	OUTCOME
Phase 1 · Now (Today)	Bug #1 (wire examples) + Bug #3 (fix Matrice rendering)	~3 hours	Grade lifts to B. Examples work, Matrice renders.
Phase 2 · Next (This week)	Bug #2 (URL-state persistence via #2 State hook)	~1 day	Grade lifts to A- . Drug list persists, is shareable.
Phase 3 · Later	Add interaction severity badges on the drug chips	~2 hours	Nice-to-have. Visual feedback before analysis.

5.2 Cross-Feature Dependencies

- ✓ Feature #2 (State Management) — the drug list persistence (Bug #2) requires the `useInteractionListState` hook from #2
- ✓ Feature #23 (Interactions Matrix) — the Matrice view IS implemented (covered in #23); Bug #3 is a rendering bug, not a missing feature
- ✓ Feature #36 (Copilot Core) — the analysis is powered by the Copilot; the drug list + patient context is sent to Gemini
- ✓ Feature #1 (Routing) — once routing ships, the URL-state persistence becomes `/interactions?drugs=8421,1523`

5.3 Final Assessment

5

EXAMPLE PAIRS

2

VIEW MODES

3

BUGS

C+

GRADE

~3h

PHASE 1 EFFORT

~1d

PHASE 2 EFFORT

FINAL VERDICT · C+ GRADE, 3 HOURS FROM B

The Interactions Checker is a **C+ grade feature**. The clinical content is excellent — the 5 example pairs are the 5 most dangerous outpatient interactions (hemorrhage, marrow aplasia, hepatotoxicity, rhabdomyolysis, hyperkalemia). The dual-view design (Vérificateur + Matrice) is correct. The ARIA tablist pattern is textbook. But three real bugs undermine the experience: example buttons don't populate the list, the drug list isn't persisted (the most expensive state-loss on the platform — a 6-drug list takes 3-5 minutes to rebuild), and the Matrice view doesn't render. Phase 1 (3 hours — wire examples + fix Matrice rendering) lifts the grade from C+ to B. Phase 2 (1 day — URL-state persistence via the #2 State hook) lifts it to A-. The clinical content is worth the investment — these 5 interactions are the ones that kill patients.

Immediate Next Steps

1. Wire the 5 quick-example buttons' onClick handlers to call setDrugs — 1 hour.
2. Debug the Matrice tabpanel conditional rendering — 2 hours.
3. Verify: click «AINS+AVK» → 2 drugs should appear in the list — 15 minutes.
4. Verify: click Matrice tab → cross-table grid should render — 15 minutes.
5. Defer Bug #2 (persistence) to Phase 2 (needs #2 State hook).

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5 dangerous example pairs, dual-view, textbook ARIA. 3 bugs (examples, persistence, Matrice). 3 hours lifts to B, 1 day lifts to A-.

5

CHAPTERS

5

SCREENSHOTS

5

EXAMPLE PAIRS

C+

GRADE
